

Synopsis

Credit Card Fraud Detection Using Machine Learning

Problem Id: 12

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- **Problem Description:**

The rapid expansion of digital payment systems, online banking, and e-commerce has brought unprecedented convenience to consumers worldwide. However, it has simultaneously created fertile ground for financial fraud. Credit card fraud alone costs the global economy over \$30 billion annually, and the volume of transactions makes manual detection practically impossible – millions of transactions occur every hour.

The core challenge lies in the extreme class imbalance inherent to fraud dataset: fewer than 0.2% of transactions are fraudulent. A naive classifier that labels every transaction as legitimate achieves over 99% accuracy yet offers zero value in practice. Effective fraud detection must therefore prioritize precision and recall over raw accuracy, correctly flagging fraud while minimizing false positives that inconvenience genuine customers.

This project aims to design and develop a complete, end-to-end machine learning classification pipeline capable of automatically identifying fraudulent credit card transactions from transaction-level financial data. The system will classify each transaction as either legitimate (Class 0) or fraudulent (Class 1) using three primary supervised learning algorithms: Logistic Regression, K-Nearest Neighbors(KNN), and Support Vector Machine (SVM), along with XGBoost as an additional high-performance model.

The pipeline encompasses data collection, preprocessing, imbalance correction via SMOTE, feature selection, model training, hyperparameter optimization using GridSearchCV, and rigorous evaluation. The outcome will be a deployable fraud detection model that assists banks and financial institutions in securing digital transactions in real time, reducing financial losses, and improving customer confidence in online payment systems.

- **Steps of Implementation:**

The project follows a structured, iterative machine learning workflow as illustrated in Figure 1:



Fig. 1 : *ML Pipeline Architecture – Credit Card Fraud Detection*

- **Data Collection:** The ULB Machine Learning Group Credit Card Fraud Dataset (Kaggle) is used, containing 284,807 real transactions made by European cardholders in September 2013. It includes 30 anonymized features (V1-V28 via PCA, plus Time and Amount) and a Class label.
- **Data Preprocessing:** Raw data is cleaned by removing duplicate records and checking for missing values. The Amount and Time features are standardized using StandardScaler. Log transformation is applied to Amount to reduce skewness, and hour-of-day is extracted from Time as an engineered feature.

- **Imbalance Handling:** Since only 0.17% of transactions are fraudulent, SMOTE (Synthetic Minority Over-sampling Technique) is applied exclusively on the training set to generate synthetic fraud samples, producing a balanced class distribution without leaking test data.
 - **Feature Selection:** The FeatureSelector library is used for automated, multi-method feature selection. It applies five complementary methods: missing values analysis, single feature importance (via gradient boosting), zero-importance features removal, low-importance features filtering, and collinear features removal. The top 20 most discriminative features are retained based on cumulative feature importance scores, reducing dimensionality, removing noise, and improving model interpretability and training efficiency.
 - **Model Training:** Four classification models are trained: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and XGBoost. An 80/20 stratified train-test split is used to preserve class proportions.
 - **Hyperparameter Tuning:** GridSearchCV with %-fold Stratified Cross-Validation is applied to each model. The scoring metric is F1 (macro), which accounts for class imbalance. Optimal parameters are selected and the best estimator is used for final evaluation.
 - **Model Evaluation:** Each tuned model is evaluated on the held-out test set using accuracy, precision, recall F1-score, confusion matrix, ROC curve, and AUC score. Models are compared to identify the strongest performer.
 - **Final Model:** The model achieving the highest F1 and AUC scores is selected as the final fraud detection system, representing the complete ML pipeline from raw data to evaluated prediction.
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- **Expected Modules/ Libraries to be used :**
 - **NumPy:** Efficient numerical computation and multi-dimensional array operations used throughout the data processing and modelling stages.
 - **Pandas:** Data loading, manipulation, cleaning, and exploratory analysis of the transaction dataset in tabular (DataFrame) form.

- **Matplotlib/ Seaborn:** Visualization library uses to plot class distributions, feature correlation heatmaps, confusion matrices, and ROC curves.
 - **Scikit-learn:** Provides the SMOTE implementation used to handle class imbalance in the training data without modifying the test set.
 - **XGBoost:** Gradient boosting framework used as an additional high-performance classifier; supports `scale_pos_weight` for native imbalance handling.
 - **FeatureSelector:** A feature selection utility built on LightGBM that automates the identification and removal of missing, zero-importance, low-importance, and collinear features. Used to select the top 20 features based on cumulative gradient boosting importance scores.
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- **Expected Evaluation metrics and Plots to be used:**
 Given the severe class imbalance, accuracy alone is insufficient. The following metrics and visualizations will be used:
 - **Accuracy:** Proportion of correctly classified transactions. Reported for completeness but not the primary metric.
 - **Precision:** Of all transactions flagged as fraud, the fraction that are genuinely fraudulent. High precision reduces false alarms.
 - **Recall (Sensitivity):** Of all actual fraud cases, the fraction correctly detected. High recall ensured minimal missed fraud.
 - **F1-Score:** Harmonic mean of precision and recall— the primary optimization metric, balancing both concerns in an imbalanced setting.
 - **Confusion Matrix:** 4-cell matrix visualizing true positives, false positive, true negatives, and false negatives for each model.
 - **ROC Curve & AUC:** Plots the true positive rate against the false positive rate at varying thresholds. AUC close to 1.0 indicates excellent distrimative ability.
 - **Required topics from the subject:**
 - **Strings:** Used for column name handling. Label encoding of categorical identifiers, printing classification reports, and managing file paths throughout the pipeline.

- **Pointers:** Python references (analogous to pointers) are leveraged when passing DataFrames and model objects between functions, enabling efficient in-place transformations without unnecessary data copying.

Files: Python file I/O operations are used to read the CSV dataset, save trained model objects using pickle/joblib, and export evaluation results and plots to disc for reporting

- **Platform Required:**

- Python 3.10+
- Jupyter Notebook (Anaconda Distribution)
- Windows 10/11

- **Books and Link Sources:**

1. Aurélien Géron — Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 3rd Edition, O'Reilly Media, 2022.
2. Christopher M. Bishop — Pattern Recognition and Machine Learning, Springer, 2006.
3. Nitesh V. Chawla et al. — SMOTE: Synthetic Minority Over-sampling Technique, Journal of Artificial Intelligence Research, 2002.
4. Kaggle Dataset — ULB Credit Card Fraud Detection:
<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
5. Scikit-learn Documentation — <https://scikit-learn.org/stable/>
6. imbalanced-learn Documentation — <https://imbalanced-learn.org/stable/>